

Neural Network - A Simple Perceptron (Theory Answers)

Q1. What is Deep Learning?

Deep Learning is a subset of machine learning that uses multi-layer neural networks to automatically learn complex patterns from data. It evolved from artificial neural networks and differs from traditional machine learning by reducing the need for manual feature engineering.

Q2. Perceptron Architecture and Limitations

A perceptron consists of input nodes, weights, a bias term, and an activation function. It computes a weighted sum of inputs and produces an output. Its main limitation is that it can only solve linearly separable problems.

Q3. Activation Functions

Activation functions introduce non-linearity into neural networks. Sigmoid outputs values between 0 and 1, Tanh outputs between -1 and 1, and ReLU outputs the input if positive and 0 otherwise. ReLU is widely used due to faster training.

Q4. Loss Function vs Cost Function

A loss function measures error for a single training example, while a cost function is the average loss across the entire dataset. Examples include Mean Squared Error and Binary Cross-Entropy.

Q5. Optimizers

Optimizers update model parameters to minimize loss. Gradient Descent uses a fixed learning rate, RMSprop adapts learning rates using moving averages, and Adam combines momentum with adaptive learning rates, making it highly effective.

Q10. Fraud Detection Workflow

Use a multilayer neural network with ReLU activations in hidden layers and Sigmoid in the output layer. Binary Cross-Entropy is suitable for binary classification. Handle class imbalance using class weights or resampling techniques. Adam is a suitable optimizer. Prevent overfitting using dropout, early stopping, and regularization. Evaluate using Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix.